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RAI TECHNOLOGY UNIVERSITY
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PROJECT REPORT

Project Title: Retail Customer Segmentation Using Purchase Behaviour.

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College of Engineering & Application

CERTIFICATE

This is to certify that **Mr. G ROSHAN ZAMEER**, bearing UEN Number **RTU24101CS035**, has successfully completed the project entitled “**Retail Customer Segmentation Using Purchase Behaviour**” under the guidance and supervision of **Mr. Abhishek S**, Department of Information Technology, during the academic year 2026–2027. The project work has been carried out by the student as part of the academic requirements of the B. Tech program, and the work presented in this report is completed satisfactorily to the best of our knowledge.

Head of Department

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Faculty Name

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ABSTRACT

The project uses the **Online Retail dataset**, which contains transaction-level information such as invoice number, product description, quantity, invoice date, unit price, customer identification, and country. The dataset initially contains **541,909 transaction records**. During data preprocessing, missing customer records, cancelled invoices, invalid quantities and invalid prices were removed. Duplicate records were also eliminated. After preprocessing, **392,692 valid transaction records** were obtained.

Customer-level features were created from the transaction data. The major features used for segmentation are **Number of Orders, Total Quantity, Total Spending, and Average Order Value**. These features represent the purchasing frequency, quantity purchased, monetary contribution, and average transaction value of each customer. A total of **4,338 unique customers** were identified after aggregation.

Three unsupervised machine learning algorithms were implemented and compared: **K-Means Clustering, Hierarchical Clustering, and DBSCAN**. StandardScaler was used to normalize the customer features before clustering. The K-Means model was developed using four clusters, while Hierarchical Clustering was also configured with four clusters. DBSCAN was tuned using different parameter combinations, with the best configuration obtained at **epsilon = 0.6 and minimum samples = 8**.

The K-Means model achieved a silhouette score of **0.9089**, Hierarchical Clustering achieved **0.9135**, and DBSCAN achieved **0.8416**. The resulting clusters were interpreted according to customer spending behaviour and categorized into segments such as **Low Value Customer, Regular Customer, High Value Customer, and Premium Customer**. A web-based application was also developed using **Flask, HTML, CSS, and JavaScript**, allowing users to enter customer purchase information and obtain segmentation results from the implemented models.

The developed system demonstrates how machine learning and web technologies can be combined to analyse retail customer behaviour and provide meaningful customer segments for business applications

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RTU24101CS035

CONTENTS

SL No.	TOPIC	Page No.
1.	Abstract	i
2.	Acknowledgement	ii
3.	Contents	iii
4.	Introduction	1
5.	Problem Statement	3
6.	Literature Survey	5
7.	Proposed Solution Methodology	7
8.	Data Collection and Processing	9
9.	Solution Development	12
10.	Result Presentation	16
11.	Discussion on Findings	18
12.	Conclusion	19
13.	References	20

CHAPTER 1

INTRODUCTION

Retail businesses interact with a large number of customers through physical stores and online platforms. Every purchase generates useful information such as the quantity purchased, product price, transaction date, invoice number, and customer identification. When this information is properly analysed, it can provide valuable insights into customer purchasing behaviour.

Customer segmentation is the process of dividing customers into different groups based on common characteristics or behaviours. Traditional customer segmentation may depend on manually defined categories or simple demographic information. However, transaction datasets can contain complex purchasing patterns that may not be easily identified using conventional methods.

Machine learning provides an effective approach for discovering hidden patterns in customer transaction data. Since customer segments are generally not predefined, **unsupervised learning** techniques can be used to automatically identify groups of customers with similar purchasing characteristics.

This project focuses on **Retail Customer Segmentation Using Purchase Behaviour**. The system analyses customer transaction data and creates meaningful customer-level features such as number of orders, total quantity purchased, total spending, and average order value.

Three clustering algorithms are implemented in this project: **K-Means Clustering, Hierarchical Clustering, and DBSCAN**. These algorithms identify customer groups based on similarities in their purchasing behaviour.

The project also includes a web-based application developed using Flask, HTML, CSS, and JavaScript. The application provides an interface where customer purchase information can be entered and the resulting customer segment can be displayed.

The main purpose of the project is to demonstrate how machine learning-based clustering can be applied to retail transaction data for identifying different types of customers and supporting data-driven business analysis.

CHAPTER 2

PROBLEM STATEMENT

Retail organizations collect large amounts of customer transaction data, but analysing this data manually can make it difficult to identify meaningful customer groups. Customers may differ significantly in terms of purchase frequency, quantity purchased, spending, and average transaction value.

2.1 Problem Definition

The objective of this project is to develop a machine learning-based system that can automatically group retail customers according to their purchasing behaviour.

Given transaction-level retail data, the system extracts customer-level features such as:

- Number of Orders
- Total Quantity
- Total Spending
- Average Order Value

These features are then used as input to clustering algorithms to identify customer segments.

2.2 Scope and Constraints

The project focuses on transaction-based customer segmentation.

The system uses the Online Retail dataset containing customer transaction information. The project considers purchasing

behaviour rather than demographic characteristics.

The segmentation is performed using unsupervised machine learning algorithms.

The project does not directly use customer age, gender, income, or other demographic information.

The quality of segmentation depends on the available transaction data and selected customer-level features.

2.3 Project Objectives

The major objectives of the project are:

- To collect and preprocess retail transaction data.
- To remove invalid, cancelled, duplicate, and incomplete transactions.
- To create meaningful customer-level purchasing features.

CHAPTER 3

LITERATURE SURVEY

Customer segmentation has been widely used in retail, marketing, e-commerce, and customer relationship management. Traditional segmentation approaches generally divide customers according to demographic characteristics or manually defined business rules. With the development of machine learning, clustering algorithms have become widely used for automatically identifying groups with similar characteristics.

3.1 Customer Segmentation Using RFM Analysis

RFM analysis is one of the commonly used approaches for understanding customer behaviour. RFM represents **Recency, Frequency, and Monetary Value**. Recency indicates how recently a customer made a purchase, frequency represents the number of purchases, and monetary value represents the amount spent.

RFM-based segmentation provides useful information about customer purchasing behaviour. However, the present project uses transaction-derived features including number of orders, total quantity, total spending, and average order value.

3.2 K-Means Clustering

K-Means is one of the most widely used unsupervised learning algorithms for customer segmentation. The algorithm divides data into a predefined number of clusters by assigning observations to the nearest cluster centroid.

The algorithm iteratively updates the cluster centroids until the assignments become stable. K-Means is computationally efficient and suitable for large customer datasets.

3.3 Hierarchical Clustering

Hierarchical Clustering creates a hierarchy of clusters by progressively merging similar observations. The resulting hierarchy can be represented using a dendrogram.

Unlike K-Means, hierarchical clustering provides a tree-like representation of relationships among observations. It can therefore be useful for examining different levels of customer grouping.

3.4 DBSCAN Clustering

DBSCAN stands for **Density-Based Spatial Clustering of Applications with Noise**. It identifies clusters based on the density of observations rather than requiring the number

of clusters to be specified in advance.

One important advantage of DBSCAN is its ability to identify noise or unusual observations. This makes it useful when customer data contains customers whose purchasing behaviour differs significantly from the majority.

3.5 Research Gap

Many customer segmentation approaches focus mainly on a single clustering algorithm. Comparing multiple clustering techniques can provide a better understanding of the structure of customer purchasing behaviour.

This project addresses this requirement by implementing and comparing **K-Means, Hierarchical Clustering, and DBSCAN** using the same customer-level features.

The project also integrates the clustering models into a web application so that segmentation results can be obtained through a user-friendly interface

CHAPTER 4

PROPOSED SOLUTION AND METHODOLOGY

4.1 Overall Architecture

The proposed system follows a modular machine learning pipeline consisting of the following stages:

Data Collection → Data Preprocessing → Feature Engineering → Customer Aggregation → Feature Scaling → Clustering → Cluster Evaluation → Customer Segment Interpretation → Web Application

The first stage involves loading the Online Retail dataset. The transaction data is then cleaned by removing missing customer records, cancelled invoices, invalid quantities, invalid prices, and duplicate records.

After preprocessing, transaction-level records are aggregated at customer level. Four important behavioural features are generated: Number of Orders, Total Quantity, Total Spending, and Average Order Value.

4.2 Implementation Strategy

The project is implemented using Python and the following major libraries:

- Pandas
- NumPy
- Scikit-learn
- Matplotlib
- Seaborn
- Joblib
- Flask
- Flask-CORS

The machine learning models are trained using the processed customer-level dataset.

The trained models and scaler are stored as model files using Joblib. The Flask backend loads these models and provides prediction and dashboard APIs.

The frontend is developed using HTML, CSS, and JavaScript. It communicates with the Flask backend and displays the customer segmentation results.

4.3 Expected Outcomes

The expected outcomes of the proposed system are:

- Identification of different customer groups.
- Detection of high-value and low-value customers.
- Identification of unusual customer behaviour using DBSCAN.
- Comparison of different clustering algorithms.
- Development of an interactive customer segmentation application.
- Presentation of customer statistics through a dashboard.

CHAPTER 5

DATA COLLECTION AND PROCESSING

5.1 Data Source

The project uses the **Online Retail dataset**, which contains transaction records from a retail business. The original dataset contains **541,909 records and 8 attributes**.

The major attributes include:

- InvoiceNo
- StockCode
- Description
- Quantity
- InvoiceDate
- UnitPrice
- CustomerID
- Country

The dataset provides transaction-level information required for analysing customer purchasing behaviour.

5.2 Data Cleaning

The raw dataset contains missing values, cancelled invoices, invalid quantities, invalid prices, and duplicate records.

The following preprocessing operations were performed:

- Records with missing CustomerID were removed.
- Cancelled invoices were removed.
- Transactions with Quantity less than or equal to zero were removed.
- Transactions with UnitPrice less than or equal to zero were removed.
- Duplicate records were removed.
- Total transaction amount was calculated using Quantity multiplied by UnitPrice.

5.3 Feature Engineering

Customer-level features were created from the cleaned transaction data.

Number of Orders

Number of unique invoices associated with a customer.

Total Quantity

Total number of products purchased by the customer.

Total Spending

Total monetary value

spent by the customer.

The transaction amount

was calculated as:

$$\text{Total Amount} = \text{Quantity} \times \text{Unit Price}$$

Average Order Value

Average amount spent by a customer per order.

$$\text{Average Order Value} = \text{Total Spending} / \text{Number of Orders}$$

These four features were selected because they represent different aspects of customer purchasing behaviour.

5.4 Feature Scaling

The customer-level features have different numerical ranges. For example, Total Spending may have values much larger than Number of Orders.

Therefore, StandardScaler was applied before clustering.

Standardization transforms the features so that they have approximately zero mean and unit variance. This prevents features with larger numerical values from dominating the clustering process.

CHAPTER 6

SOLUTION DEVELOPMENT

6.1 K-Means Clustering

K-Means clustering was implemented to divide customers into four groups.

The algorithm assigns each customer to the nearest cluster centroid and repeatedly updates the centroids until the clustering becomes stable.

The K-Means model generated

four customer clusters. The

obtained silhouette score was:

Silhouette Score = 0.9089

The clusters were interpreted based on their

spending characteristics. The identified customer

categories were:

- Low Value Customer
- Regular Customer
- High Value Customer
- Premium Customer

6.2 Hierarchical Clustering

Hierarchical Clustering was implemented using four clusters.

The algorithm groups customers according to their similarity and creates a

hierarchical structure. The obtained silhouette score was:

Silhouette Score = 0.9135

The resulting clusters were interpreted using customer spending behaviour and mapped to customer categories.

6.3 DBSCAN Clustering

DBSCAN was implemented to identify dense customer groups and unusual observations. Parameter tuning was performed to identify suitable values for epsilon and minimum samples. The best configuration obtained

Model Comparison

The clustering models were evaluated using the silhouette score.

Algorithm	Number of Clusters	Silhouette Score
K-Means	4	0.9089
Hierarchical Clustering	4	0.9135
DBSCAN	2	0.8416

The silhouette score measures how well observations fit within their assigned clusters compared with other clusters.

6.1 Web Application Development

A web-based application was developed to make the segmentation system accessible through a user interface. The backend was implemented using **Flask**.

The frontend was developed using:

- HTML
- CSS
- JavaScript
- Chart.js

The application contains a customer input form where the user can enter:

- Number of Orders
- Total Quantity
- Total Spending
- Average Order Value

The entered values are passed to the Flask backend.

The backend applies the saved scaler and clustering models and returns the segmentation results. The application displays:

- K-Means cluster
- Hierarchical cluster
- DBSCAN cluster
- Customer segment
- Customer behaviour analysis

6.2 Dashboard Development

The application also contains an interactive dashboard. The

dashboard displays:

- Total Customers
- Total Orders
- Total Quantity
- Total Spending
- Average Order Value
- K-Means Cluster Distribution
- Hierarchical Cluster Distribution
- DBSCAN Cluster Distribution
- Silhouette Scores

This allows users to understand the overall customer dataset and clustering results.

CHAPTER 7

RESULT PRESENTATION

7.1 Dataset and Preprocessing Results

The Online Retail dataset was successfully loaded and processed using Python. The original dataset contained **541,909 transaction records and 8 attributes**. During preprocessing, records with missing CustomerID, cancelled invoices, invalid quantities, invalid unit prices, and duplicate records were removed.

After preprocessing, **392,692 valid transaction records** remained. The cleaned transaction data was then used for customer-level analysis.

```
=====
RETAIL CUSTOMER SEGMENTATION
=====

Loading dataset...
Original dataset shape: (541909, 8)

Preprocessing data...
After preprocessing: (392692, 8)

Creating customer-level features...

Number of customers: 4338

Features used:
- NumberOfOrders
- TotalQuantity
- TotalSpending
- AverageOrderValue
```

7.2 Customer Feature Generation

After preprocessing, the transaction records were grouped according to CustomerID to create customer-level purchasing features.

The following four features were generated:

- **Number of Orders**
- **Total Quantity**
- **Total Spending**
- **Average Order Value**

A total of **4,338 customers** were identified from the processed dataset. These features were used as input variables for the clustering algorithms.

```
=====
CUSTOMER LEVEL FEATURES
=====
```

CustomerID	NumberOfOrders	TotalQuantity	TotalSpending	AverageOrderValue	KMeans_Cluster	Hierarchical_Cluster	DBSCAN_Cluster
12346.0	1	74215	77183.60	77183.600000	1	2	-1
12347.0	7	2458	4310.00	615.714286	0	0	0
12348.0	4	2341	1797.24	449.310000	0	0	0
12349.0	1	631	1757.55	1757.550000	0	0	0
12350.0	1	197	334.40	334.400000	0	0	0
12352.0	8	536	2506.04	313.255000	0	0	0
12353.0	1	20	89.00	89.000000	0	0	0
12354.0	1	530	1079.40	1079.400000	0	0	0
12355.0	1	240	459.40	459.400000	0	0	0
12356.0	3	1591	2811.43	937.143333	0	0	0

7.3 Clustering Results

Three unsupervised machine learning algorithms were implemented for customer segmentation: **K-Means Clustering, Hierarchical Clustering, and DBSCAN**.

K-Means and Hierarchical Clustering were configured to generate four clusters. DBSCAN was tuned using different parameter combinations and the selected configuration used an epsilon value of **0.6** and minimum samples of **8**.

The clustering algorithms grouped customers according to similarities in their purchasing behaviour.

```
=====
K-MEANS CLUSTERING
=====
Number of clusters: 4
Silhouette Score: 0.9089

=====
HIERARCHICAL CLUSTERING
=====
Number of clusters: 4
Silhouette Score: 0.9135

=====
DBSCAN CLUSTERING
=====

Searching for suitable DBSCAN parameters...

Best EPS: 0.6
Best Min Samples: 8
Number of Clusters: 2
Noise Points: 82
Noise Percentage: 1.89 %
Silhouette Score: 0.8416
```

7.4 Model Performance Comparison

The performance of the clustering algorithms was evaluated using the **Silhouette Score**. The silhouette score measures the quality of the clusters by considering the similarity of customers within their assigned cluster and their separation from other clusters.

The obtained scores were:

Algorithm	Number of Clusters	Silhouette Score
K-Means	4	0.9089
Hierarchical Clustering	4	0.9135
DBSCAN	2	0.8416

7.5 Customer Segment Prediction

The trained clustering models were integrated with the Flask backend of the web application. The user can enter customer purchasing information such as number of orders, total quantity, total spending, and average order value.

The entered information is processed using the saved scaler and clustering models. The application then displays the clustering results and corresponding customer segment.

The screenshot shows a web application interface for predicting customer segments. It features a dark blue background with white and light blue text and buttons.

Customer Purchase Information
Enter customer purchase details to identify the customer segment using machine learning.

Number of Orders: 10
Total Quantity: 100
Total Spending: 1000
Average Order Value: 200

Predict Customer Segment (button) **Clear** (button)

Model Predictions

- K-Means**: Low Value Customer (Cluster 1)
- Hierarchical Clustering**: Low Value Customer (Cluster 1)
- DBSCAN**: Low Value Customer (Cluster 1)

Final Customer Segment: Low Value Customer

7.6 Web Application and Dashboard

A web-based customer segmentation application was developed using **Flask, HTML, CSS, JavaScript, and Chart.js**.

The application provides an interface for entering customer information and displaying segmentation results. It also contains a dashboard showing important customer statistics and clustering information.



CHAPTER 8

DISCUSSION ON FINDINGS

8.1 Customer Segmentation Findings

The clustering analysis demonstrates that customers exhibit different purchasing behaviours.

Some customers have relatively low purchase activity, while others place more orders and contribute greater spending. The customer-level features allow these differences to be represented numerically.

K-Means divides customers into four behaviour-based groups, which are interpreted as Low Value, Regular, High Value, and Premium customers.

8.2 Comparison of Clustering Algorithms

The three clustering algorithms produced different cluster structures.

K-Means generated four customer groups and achieved a silhouette score of 0.9089. Hierarchical Clustering also generated four groups and achieved a silhouette score of 0.9135.

DBSCAN generated two clusters and identified 82 observations as noise, with a silhouette score of 0.8416. The differences occur because each algorithm uses a different clustering approach.

K-Means uses cluster centroids, Hierarchical Clustering builds groups based on hierarchical similarity, while DBSCAN uses density-based grouping.

8.3 Interpretation of Customer Segments

Low Value Customer

These customers generally demonstrate relatively lower purchase activity and spending.

They may require suitable promotional campaigns and engagement strategies to increase purchase frequency.

Regular Customer

These customers demonstrate consistent purchasing activity and contribute regular revenue to

the business. Maintaining customer engagement and providing relevant offers can help retain this segment.

High Value Customer

These customers demonstrate higher purchasing activity or spending compared with regular customers. They represent an important customer group for targeted marketing and retention strategies.

Premium Customer

Premium customers represent customers with particularly strong purchasing behaviour and monetary contribution.

Businesses can provide personalized services, loyalty benefits, and targeted offers to this segment.

8.1 DBSCAN Noise Analysis

DBSCAN identified 82 customers as noise points.

These observations do not belong strongly to the dense clusters identified by the algorithm.

Such customers may represent unusual purchasing patterns, extremely high-value customers, or customers with behaviour that differs from the majority.

Therefore, DBSCAN provides an additional perspective by identifying potentially unusual customer behaviour.

8.2 Web Application Findings

The developed web application successfully integrates the trained machine learning models with a user interface.

The user can enter customer purchasing information and receive clustering results.

The dashboard provides a visual representation of customer statistics and clustering performance.

CHAPTER 9

CONCLUSION

This project developed a machine learning-based system for Retail Customer Segmentation Using Purchase Behaviour. The Online Retail dataset containing 541,909 transaction records was successfully processed to remove missing, cancelled, invalid, and duplicate transactions. After preprocessing, 392,692 valid transaction records were obtained. Customer-level aggregation resulted in 4,338 customers. Four important behavioural features were generated: Number of Orders, Total Quantity, Total Spending, and Average Order Value. Three unsupervised machine learning algorithms were implemented: K-Means Clustering, Hierarchical Clustering, and DBSCAN. K-Means generated four clusters with a silhouette score of 0.9089. Hierarchical Clustering generated four clusters with a silhouette score of 0.9135. DBSCAN generated two clusters and identified 82 noise points, achieving a silhouette score of 0.8416. The clusters were interpreted according to customer purchasing behaviour and categorized into Low Value Customer, Regular Customer, High Value Customer, and Premium Customer segments where applicable. A web-based application was also developed using Flask, HTML, CSS, JavaScript, and Chart.js. The application allows users to enter customer purchasing information and obtain segmentation results from the trained models. The dashboard provides overall customer statistics and visual representations of the clustering results. The project demonstrates that unsupervised machine learning can be applied to retail transaction data to identify meaningful customer groups. Such segmentation can provide useful information for customer analysis, targeted marketing, customer retention, and business decision-making. Future work can extend the system by incorporating additional customer features, product-level information, purchase recency, demographic information, real-time transaction data, and advanced clustering techniques. The web application can also be enhanced with authentication, database integration, downloadable reports, and deployment to a cloud platform.

CHAPTER 10

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